# NAVEENRAJ S.S

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ELECTRONICS & COMMUNICATION ENGINEER · EMBEDDED SYSTEMS & PCB DESIGNER

## CORE SKILLS

- Embedded C / C++
- ESP32 & Arduino
- IoT System Design
- PCB Design
- Embedded Automation
- Sensor Integration
- Python & AI/ML (TinyML)
- UI / Web Design

## TECH STACK

**Microcontrollers**
ESP32 · ARM Cortex-M · Arduino

**Hardware & Analog**
Op-Amps · Rectifier & Filter Design . RF Modules · FPGA

**PCB Design**
KiCad · Altium

**Protocols & Wireless**
UART · BLE · Wi-Fi ESP-NOW · SPI · I²C

**AI / Vision**
Signal Processing · Computer Vision · YOLOv5 · OpenCV · Machine Learning

**Languages**
Embedded C · C++ · Python

**UI / Web Design**
React · Tailwind CSS Framer Motion ·HTML5 . CSS3

## PROFILE

3rd-year ECE student with hands-on experience in embedded systems and PCB design. Designed PCBs from schematic to Gerber output using KiCad, Altium, and developed IoT and AI-based embedded projects using ESP32, Embedded C, and TensorFlow Lite. Passionate about R&D in embedded systems and looking for an internship to apply these skills in a real-world engineering environment.

## EDUCATION

**B.E. — Electronics & Communication Engineering**
**KCG College of Technology, Chennai** · Sep 2024 – Present

Sem 4 · CGPA: 8.15 / 10.00

## PROFESSIONAL EXPERIENCE

**Project Intern — Healthcare Technology & Entrepreneurship**
**Ophthall Eye Care Practice Management, Chennai** · 2 Month · (1st Sept- 31st Oct)2026

- Explored ophthalmic diagnostic and surgical equipment, focusing on their technical features, applications, and role in modern eye-care technology.
- Gained exposure to medical-device ecosystems and the intersection of electronics, imaging technology, product evaluation, and healthcare applications.

Reference: https://www.ophthall.in/

**Industrial Trainee — Data Networking**
**Bharat Sanchar Nigam Limited (BSNL), Chennai** · 1 Week · 2025

- Completed structured training on routing protocols, switching architectures, and enterprise communication systems.
- Gained firsthand exposure to large-scale telecom infrastructure and industrial networking setups.

## KEY PROJECTS

### RF Defence System — PCB Design & Embedded | *Jan – Mar 2026*

*Stack: KiCad · ESP32 · LM358 Op-Amp · Embedded C*

- Designed a full RF interference detector PCB from schematic to Gerber/DRL outputs using KiCad; integrated LM358 Op-Amp with ESP32 to detect and directionally track anomalous wireless signals.

**GitHub:** github.com/NAVEENKCG/RF_Defence_System_PCB

### Smart Traffic Control — IoT & Embedded Automation | *Jul – Sep 2025*

*Stack: Python · YOLOv5 · OpenCV · ESP32*

- AI-driven embedded system using YOLOv5 for real-time vehicle detection; ESP32 GPIO controls relay-switched signal timings dynamically based on live traffic density.

**GitHub:** github.com/NAVEENKCG/Smart_Traffic_ctrl

## CERTIFICATIONS

FPGA Architecture
GUVI / HCL

IoT Boot Camp
KCG Centre of Excellence

Data Networking
BSNL, Chennai

Arduino Programming
GUVI / HCL

## ACHIEVEMENTS

**Coordinator — Young Technocrats 3.0**
Led KCG-wide innovation summit

**Hackathonix '26**
Built BookBridge (full-stack) in 48 hrs

**SCIFEST '25 — Ideathon**
KCG Innovation Council, National Science Day

**IoT Boot Camp**
Accident Rescue System — KCG ECE CoE

### Posture Detection & Thrombosis Prevention — IoT Sensor System |

*Nov 2025 – Feb 2026*

*Stack: Sensors · Microcontroller · Embedded C · Python*

- Real-time posture monitoring system using multiple sensors and a microcontroller; sensor data is classified on-device and alerts are pushed wirelessly to the user within < 100 ms.

**GitHub:** github.com/NAVEENKCG/Posture_Detection

### Orbit AI — Mind-Controlled Assistive System (BCI) | *In Development*

*Stack: ESP32 · TGAM · TensorFlow Lite · BLE · Wi-Fi · ESP-NOW*

- Wearable non-invasive BCI: EEG signals acquired via TGAM, classified on-device using TinyML on ESP32/ARM Cortex-M, and delivered over BLE/Wi-Fi/ESP-NOW with < 50 ms latency to control wheelchairs and smart home devices.

**GitHub:** github.com/NAVEENKCG/Mind_Wave_AImodel

## WEB DESIGN

Designed and built responsive UI layouts using React, Tailwind CSS, and Framer Motion animations:

- MindWave Web — UI dashboard for BCI system data visualisation.
- BookBridge — Full-stack book-sharing platform designed and built in 48 hours at Hackathonix '26.
- Portfolio Sites — Personal and client portfolios with animated, responsive layouts.

**GitHub:** github.com/NAVEENKCG